Collimated Displaced e^+e^- Pairs from Light ALPs in ttbar Events: a Toy-MC Look at Magnetic-Field Separation and Photon-Conversion Backgrounds

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Companion to: J. Alimena, F. Blekman, C. Cazzaniga, X. Cid Vidal, J. Niedziela, L. Rygaard, R. Soto Rey, *Hunting for Long-Lived ALPs in Top Events via Displaced Dielectron Pairs at the LHC*, arXiv:2610.38932. Code: github.com/wspaik5/ALP_ttbar_Displaced_Dielectron_Toy_MC_arXiv2610.38932

Disclaimer. This is a simplified, numpy-only toy Monte Carlo written out of curiosity after reading the paper above. It is not affiliated with the authors or with the CMS Collaboration, uses parametrised ALP kinematics and assumed detector numbers, and should be read as qualitative trends only. The paper itself explicitly places detector-level reconstruction and instrumental backgrounds outside its scope; this note simply explores two of those effects in a rough way.

1. Motivation

The paper extends the top-associated displaced-lepton strategy to $a \to e^+e^-$, reaching ALP masses down to 17 MeV, below the dimuon threshold. Two challenges flagged by the authors for the lowest masses are (i) the very small e^+e^- opening angle, evaluated in the paper with straight-line trajectories, and (ii) instrumental backgrounds such as interactions with detector material. The authors note that the opposite bending of e^+ and e^- in the CMS field may mitigate (i). This note quantifies (i) with and without the field and gives a first rough look at (ii) via photon conversions.

2. Toy model

ALP kinematics. $p_T(a) = 20\ \text{GeV} + \text{Exp}(80\ \text{GeV})$, $\eta \sim N(0, 1.6)$; exponential decay with lab length $\beta\gamma c\tau$; isotropic two-body decay boosted to the lab. The parametrisation was tuned so that the *shape* of the pre-selection efficiency over $(m_a, c\tau)$ matches Table 3 of the paper (rms log-ratio 0.26); the absolute toy efficiency is a factor ~3.6 higher than the paper (unresolved; likely from effects not modelled, e.g. FSR and the full ttbar+a event).

Detector (barrel only, $|\eta| < 1.2$). 14 CMS Phase-1 barrel layer radii (4 pixel, 4 TIB, 6 TOB); uniform 3.8 T solenoid with exact helix–cylinder intersections; a pair is "resolved" if its two hits are separated by more than an assumed two-hit resolution (0.2 mm pixel, 0.4 mm strip) in at least 3 layers. Pre-selection follows the paper: 0.5 mm $< l_{xy} <$ 0.8 m.

Conversions. Isolated photons from ttbar+γ with an assumed $\sigma(p_T^{\gamma} > 20\ \text{GeV}) \approx 1$ pb, p_T^{-3} spectrum, 500 fb^{-1}; an assumed material map (beam pipe, pixel layers and supports, strip layers; $\approx 0.46\ X_0$ at $\eta = 0$, conversion probability $7/9 \cdot x/X_0$); Bethe–Heitler-like energy sharing. Track directions are smeared (0.3 mrad pixel-seeded / 0.8 mrad strip-only, plus 2 mrad·GeV/p multiple scattering, 2% momentum resolution) and vertex radii by 1 mm (pixel region) / 3 mm. Cuts mirror the paper (min. electron $p_T > 1$ GeV) plus pair $p_T > 20$ GeV. Expected limits use a crude counting rule $N_{up} = \max(3, 2\sqrt{B})$, so only *ratios* are meaningful.

3. Results

3.1 Geometry and validation. Figure 1 shows the decay-position and boost effects behind the paper's efficiency pattern, and the toy-vs-paper comparison of pre-selection efficiencies (after one global normalisation factor).

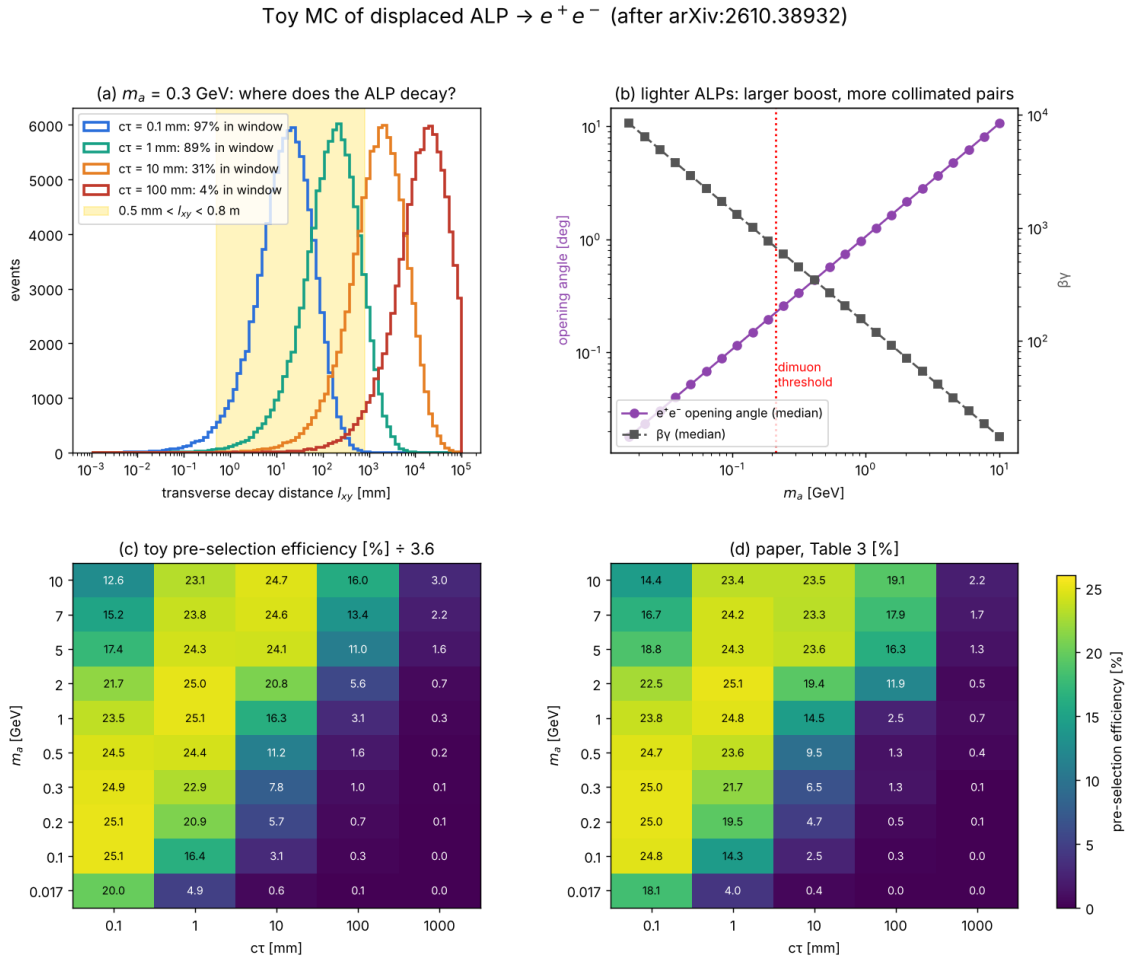


Figure 1. (a) Decay-distance distribution for m_a = 0.3 GeV; (b) median opening angle and boost vs. mass; (c) toy pre-selection efficiency divided by 3.6; (d) paper Table 3.

3.2 Magnetic-field separation. With straight lines, a 17 MeV pair stays below the assumed two-hit resolution in most layers; in the 3.8 T field the e^+ and e^- bend apart and reach centimetre separations by the outer tracker.

m_a	straight lines	3.8 T field
17 MeV	55%	97%
40 MeV	82%	97%
100 MeV	95%	97%
300 MeV	99.5%	98.6%

Table 1. Fraction of pre-selected pairs resolved in ≥ 3 barrel layers (cτ = 1 mm; results for cτ = 0.1 and 10 mm are within a few percent). Above ~0.2 GeV the field costs 1–2% because the softest electrons curl before reaching three layers.

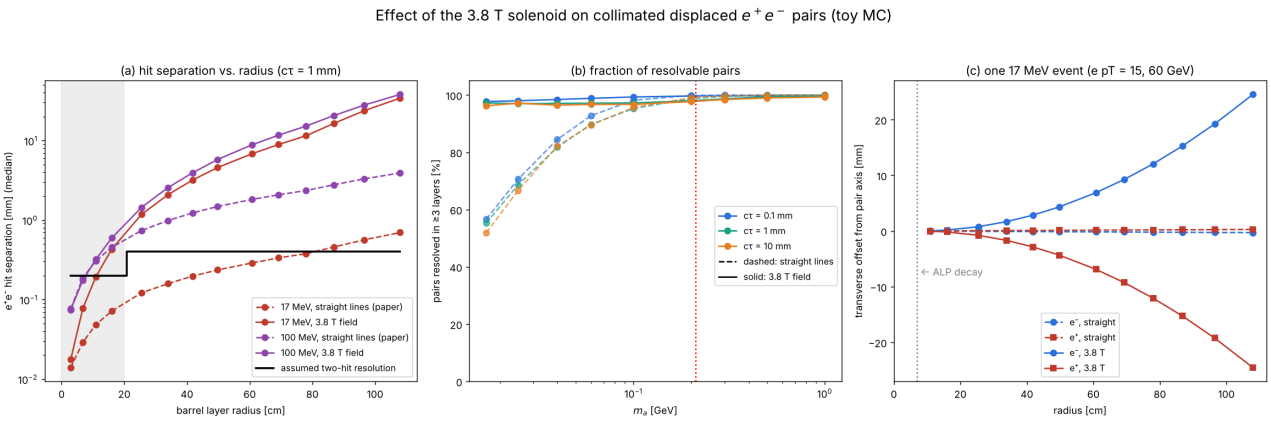


Figure 2. (a) Median hit separation per layer; (b) resolvable fraction vs. mass for three lifetimes; (c) one 17 MeV event.

3.3 Photon conversions. Under the assumptions above, about 9×10^{4} conversions pass the paper-like cuts in the barrel, compared with tens of signal events at $\sigma\times B \sim 10^{-3}$ pb. Conversions cluster at material radii, whereas ALP decays are spread smoothly; the measured conversion mass is resolution-dominated and overlaps a 17 MeV signal, but not 100–200 MeV signals.

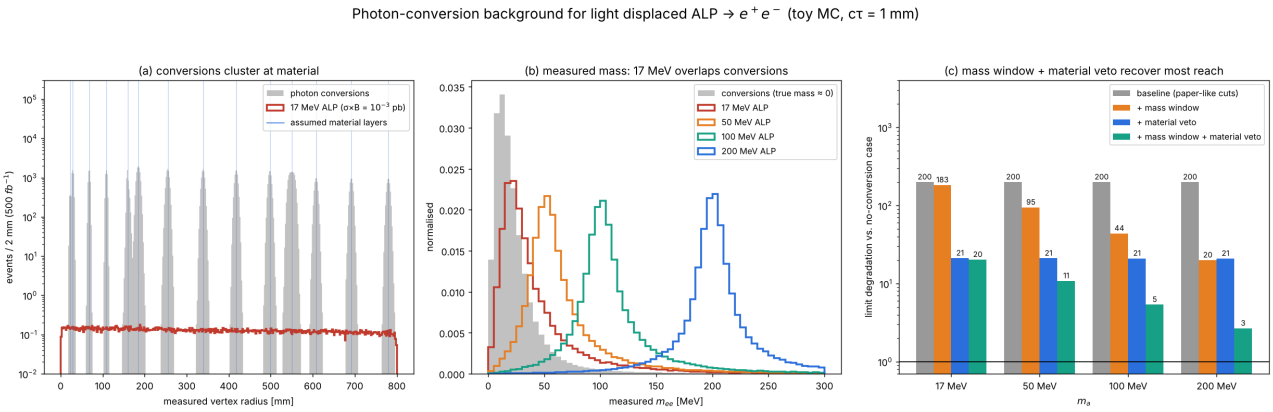


Figure 3. (a) Measured vertex radius; (b) measured m_{ee}; (c) degradation of the expected limit relative to the no-conversion case for successive selections (cτ = 1 mm).

m_a	baseline	+ mass window	+ material veto	both
17 MeV	200x	183x	21x	20x
50 MeV	200x	95x	21x	11x
100 MeV	200x	44x	21x	5x
200 MeV	200x	20x	21x	3x

Table 2. Factor by which the expected limit worsens because of conversions (toy, crude counting rule).

4. Takeaways

• The B field, neglected in the paper's straight-line treatment, substantially resolves the collimation issue at the lowest masses in this toy, consistent with the authors' remark.

• Photon conversions could be a dominant background in the sub-100 MeV region. At 17 MeV the mass does not discriminate, so a material-position veto (which itself relies on field-assisted vertexing of nearly parallel tracks) appears to be the key handle.

• The relevant quantities for a realistic estimate are the material map, the conversion-vertex resolution, and the reconstruction efficiency for displaced, collimated tracks, all of which require detector-level simulation.

5. Limitations

Parametrised (not generated) ALP and photon kinematics; barrel only; no FSR, pile-up, track-finding efficiency, calorimeter or isolation modelling; assumed material budget and resolutions; ttbar+γ rate is an order-of-magnitude assumption; other backgrounds from the paper (e.g. heavy-flavour decays) are not included; the factor ~3.6 normalisation offset relative to the paper is not understood. Absolute numbers may be off by large factors; trends are the intended content.

6. Reproducibility

All figures and numbers are reproduced by `./run_all.sh` (Python 3, numpy, matplotlib; ~30 s). Repository: github.com/wspaik5/ALP_ttbar_Displaced_Dielectron_Toy_MC_arXiv2610.38932. Fixed random seeds are used throughout.